2 (2026) 100215

Contents lists available at ScienceDirect

Strategic Business Research

journal homepage: www.sciencedirect.com/journal/strategic-business-research

Digital resilience and continuous use of health information systems among healthcare workers: A systematic literature review

Mudasir Issah *, Ibrahim Osman Adam

Department of Business Analytics and MIS, School of Business, University for Development Studies, Tamale, Ghana

A R T I C L E I N F O

Keywords:
Digital resilience
Health information systems
Continuous use
Technostress
Healthcare workers
Systematic literature review

A B S T R A C T

The increasing digitalization of healthcare has intensified the need for healthcare workers to continuously engage with health information systems (HIS) such as electronic health records, telemedicine platforms, and clinical decision support tools. However, challenges related to digital disruptions, technostress, and system discontinuance continue to undermine the effectiveness of these technologies. Despite growing scholarly interest, the literature on healthcare workers' digital resilience and continuous HIS use remains fragmented, theoretically underdeveloped, and methodologically limited. This study addresses this gap by conducting a systematic literature review combined with a grounded theory approach to synthesize existing research, identify key themes, methodological trends, and theoretical foundations, and propose a framework for future inquiry. Following established review protocols, a total of 60 peer-reviewed studies comprising 53 papers retrieved from four major databases (Scopus, AIS eLibrary, PubMed, and IEEE Xplore) and 7 papers identified through backward and forward citation tracking were subjected to rigorous inclusion and exclusion criteria. The findings reveal that existing research clusters around four core themes: (1) antecedents and enablers of digital resilience and HIS use, (2) digital threats and technostress creators, (3) coping mechanisms and resilient behaviors, and (4) outcomes of digital resilience and continuous HIS use. Methodologically, the literature is dominated by cross-sectional surveys and qualitative interviews, with a notable absence of longitudinal, action research, and objective system-use studies. Theoretically, nearly half of the reviewed studies lack explicit theoretical grounding, while those that do rely predominantly on borrowed frameworks such as the Job Demands-Resources model, the Technology Acceptance Model, and UTAUT. Based on these findings, this study proposes a framework that maps the current state of knowledge and charts an agenda for future research, calling for domain-specific theory development, longitudinal methodological designs, and deeper investigation into role-specific resilience dynamics and the implications of emerging technologies such as artificial intelligence. This review contributes to the information systems and health informatics literature by consolidating a rapidly growing but dispersed field and providing researchers and practitioners with gaps for advancing future healthcare digital resilience research.

Introduction

The increasing digitalization of healthcare has transformed how healthcare workers work to deliver, document, and coordinate care. Over the past decade, the introduction of Health Information Systems (HIS) such as electronic health records (EHRs), telemedicine platforms, and clinical decision support tools has enabled more efficient and data-driven care processes (Janssen et al., 2022). However, these technologies have also introduced new challenges for the healthcare workforce, particularly concerning adaptation, sustainability, and resilience in digital work environments (Al-Anezi, 2025). As healthcare systems undergo rapid technological evolution, the ability of healthcare workers to adapt to disruptions and sustain continuous use of digital tools has become critical for both individual and organizational performance (Salifu Sidii, 2024).

Digital resilience, in the healthcare worker context, refers to the capacity of healthcare professionals to recover from digital disruptions, adapt to new technological demands, and maintain effective performance despite technological change or failure (Liu et al., 2023). It encompasses psychological, behavioral, and technical dimensions that enable continuity of care during system downtimes, updates, or cybersecurity events (Liu et al., 2023).

* Corresponding authors at : University for Development Studies.
E-mail address: mudasiru.issah@uds.edu.gh (M. Issah).

https://doi.org/10.1016/j.sbr.2026.100215
Received 4 April 2026; Received in revised form 23 June 2026; Accepted 25 June 2026
Available online 27 June 2026

At its core, digital resilience in healthcare is a socio-technical construct, reflecting the interdependence between people and technology. The socio-technical systems theory provides a useful lens for understanding this relationship, as it posits that effective performance results from the joint optimization of social and technical subsystems (Bostrom and Heinen, 1977; Hariyanto et al., 2018; Sittig and Singh, 2010).

It is important to delineate digital resilience from a set of related but conceptually distinct constructs that are often used interchangeably in the literature. Digital resilience differs from technology adoption, which refers to the initial decision to accept or use a system (Davis, 1989; Venkatesh et al., 2003) and is typically a one-time event. It also differs from continuance intention, which is the user's post-acceptance intention to keep using a system (Bhattacherjee, 2001); continuance is a behavioural outcome, whereas resilience is the underlying adaptive capacity that produces it under disruption. Digital resilience similarly extends beyond IT-enabled coping, which describes specific responses to particular techno-stressors (Pirkkalainen et al., 2019), by encompassing the broader anticipatory, monitoring, response, and learning capabilities (Hollnagel et al., 2017) that enable healthcare workers to function across multiple, evolving digital disruptions. Finally, digital resilience must be distinguished from general psychological resilience, which is a stable personal trait or process of bouncing back from adversity (Luthans, 2002); digital resilience is domain-specific, situated, and shaped by the sociotechnical configuration of healthcare work.

Digital resilience in healthcare is fundamentally a socio-technical phenomenon. Drawing on socio-technical systems theory, effective performance depends on the joint optimization of social and technical subsystems (Bostrom and Heinen, 1977; Hariyanto et al., 2018; Sittig and Singh, 2010). Healthcare workers' capacity to sustain HIS use during disruptions is influenced not only by individual capabilities but also by organisational support, leadership, training, and collaborative practices. Likewise, technical factors such as system usability, interoperability, and reliability shape users' ability to adapt to technological change and recover from disruptions.

In practice, health information systems (HIS) have become integral to modern healthcare delivery, requiring healthcare workers to continuously interact with digital technologies in their daily clinical activities. Governments and healthcare institutions worldwide continue to promote digital transformation initiatives aimed at strengthening service delivery and resilience in healthcare systems. However, despite these investments, challenges related to healthcare workers' adaptation, digital stress, and discontinuance of system use persist. Therefore, to advance research on healthcare workers' digital resilience and continuous use of HIS, it is important to systematically review the extant literature to unearth: (1) extensively researched issues, (2) less-researched areas, (3) methodologies and theories employed in prior studies, as well as (4) critical gaps that warrant future investigation.

In the extant information systems (IS) and healthcare literature, limited attempts have been made to systematically review studies that examine healthcare workers' digital resilience in relation to the continued use of HIS. While prior research has explored HIS adoption, user satisfaction, and post-implementation outcomes, there remains a fragmented understanding of how digital resilience enables continued use of HIS. Thus, we argue that a comprehensive review of existing literature in this domain will: (1) enable a structured evaluation of the extent and focus of studies already undertaken and identify gaps for future research; and (2) provide a clearer conceptual understanding of digital resilience as a critical determinant of continuous HIS use among healthcare workers.

Therefore, this study seeks to offer a critical review of research on healthcare workers' digital resilience and HIS continuance by synthesizing existing studies and developing a framework that highlights key issues, methodological approaches, and theoretical foundations underpinning prior investigations. Thus, the study is motivated by the following research questions:

RQ1: What themes have been investigated in prior research on healthcare workers' digital resilience and continuous use of health information systems?
RQ2: What methodologies and methods have been employed in these studies?
RQ3: What theories, models, and frameworks have informed existing research?
RQ4: How can the identified thematic, methodological, and theoretical gaps be integrated into a conceptual framework to guide future empirical research?

While RQ1–RQ3 provide the descriptive foundation of the systematic review (i.e., themes, methods, and theories), RQ4 provides the integrative synthesis. It moves beyond cataloguing what has been studied toward identifying conceptual blind spots, methodological deficits, and theoretical gaps. RQ4 is therefore the question whose answer is operationalised through an integrative conceptual framework (Fig. 3) and a set of testable propositions that map out a forward-looking future research agenda rather than retrospective classification alone.

To position this study against prior reviews that have addressed adjacent topics in the digital health and healthcare worker IS-use literature, Table S2 (see supplementary material) provides a comparative summary of the scope, sample size, methodology, focal constructs, and contribution of this review relative to four recent reviews of comparable scope (Ang et al., 2022; Larsson et al., 2025; Sun et al., 2022; Woldemariam and Jimma, 2023). Our review is distinctive in three respects. First, it is the only review to date that jointly examines digital resilience and continuous HIS use as interlocking constructs, rather than studying them in isolation. Second, it integrates a grounded-theory coding logic with the systematic review protocol, enabling both rigor and conceptual development. Third, it explicitly maps thematic, methodological, and theoretical gaps onto a single integrative framework that can guide future research agenda.

Methods

Following established systematic literature review protocols (Kitchenham, 2004; Tranfield et al., 2003), this study employed a systematic literature review approach combined with the grounded theory literature review method proposed by Wolfswinkel et al. (2013). While the systematic review provides a structured procedure for identifying, refining, and analyzing relevant studies, the grounded theory approach ensures the extraction of linkages between different papers through iterative coding and conceptualization. The two methods were combined to achieve a comprehensive literature coverage and rigorous analytical depth. In line with these approaches, this study followed a five-stage systematic review process: (1) definition of literature inclusion and exclusion criteria, (2) literature search, (3) literature refinement, (4) analysis of selected literature, and (5) presentation of findings. Ultimately, this process yielded a final corpus of 60 peer-reviewed studies, 53 studies extracted from primary electronic databases and 7 studies recovered through citation tracking. These steps are discussed in the below sections.

Definition of literature inclusion/exclusion criteria

To ensure the quality of the review, inclusion and exclusion criteria were clearly defined prior to conducting the literature search. Consistent with Webster and Watson (2002), the review primarily targeted peer-reviewed journal articles and conference papers. The review targeted studies that explicitly examined digital resilience, continuous use of health information systems (HIS), or related constructs such as technology continuance, post-adoption behavior, and user adaptation within healthcare settings. No specific date restriction was applied to the search, in order to ensure comprehensive coverage of relevant literature across the development of the field (Jesson et al., 2011; Sauer and

Seuring, 2023). The final literature search was conducted in January 2026. Search terms included "digital resilience," "healthcare workers," "health information systems," "continuous use," "technology continuance," and "post-adoption," with Boolean operators used to combine and refine results.

To enhance the transparency and reproducibility of the literature search, the full Boolean query strings used for each database are presented in supplementary material Table S1. The queries were adapted to the syntactic conventions of each database while preserving the same conceptual structure: a digital resilience construct combined with a healthcare worker construct and a continuous use or post-adoption construct. For example the scopus search string was: *TITLE-ABS-KEY (("digital resilien*" OR "technostress" OR "technology resilien*" OR "IT resilien*") AND ("health information system*" OR "electronic health record*" OR "EHR" OR "EMR" OR "telemedicine" OR "mHealth" OR "clinical decision support" OR "hospital information system*") AND ("healthcare worker*" OR "nurse*" OR "physician*" OR "clinician*" OR "health professional*" OR "medical staff") AND ("continuous use" OR "continuance" OR "post-adoption" OR "sustained use" OR "continued use" OR "coping" OR "adaptation" OR "workaround*"))*. The asterisk (*) was used as a truncation operator to capture morphological variants (e.g., "resilien*" retrieves "resilience," "resilient," and "resiliency"). Filters were applied to restrict results to peer-reviewed journal articles and conference proceedings published in English.

In addition to the database-specific searches, manual backward and forward citation tracking was performed on highly cited and recent articles in the field. To minimise selection bias, two reviewers independently screened titles and abstracts of all retrieved records against the inclusion and exclusion criteria. Initial inter-rater agreement at the title-and-abstract stage was 87.4% (Cohen's $\kappa = 0.78$, indicating substantial agreement), and all disagreements were resolved through discussion and, where consensus could not be reached, by consultation with a neutral reviewer. Full-text screening was performed jointly through consensus-based review thus, an inter-rater reliability statistic was not calculated for the full-text eligibility stage. While this approach ensured consistent interpretation of the inclusion criteria, it represents a limitation of the review process.

Literature search

The literature search was conducted systematically across four databases: Scopus, AIS eLibrary, PubMed, and IEEE Xplore. These databases were selected because they collectively capture a broad range of publications in both healthcare and information systems research. The searches were performed using the predefined keywords in article titles, abstracts, and keywords. To ensure completeness, both forward and backward searches were undertaken. In the backward search, the reference lists of the initially retrieved articles were manually reviewed to identify additional relevant studies. In the forward search, the Google Scholar citation function was used to trace more recent studies that cited the initially identified articles. This multi-step search process resulted in the retrieval of 1639 peer-reviewed papers for initial consideration.

Literature refinement

Following the search stage, the retrieved studies were subjected to a refinement process to remove duplicates and irrelevant papers. Each article title, abstract and full text was reviewed to determine its relevance to the research objectives. Studies that mentioned digital resilience or HIS use only in passing, without substantial conceptual or empirical discussion, were excluded. Similarly, articles that focused solely on system design or technical architecture without reference to healthcare workers' usage or behavioral aspects were omitted. After this process, 60 studies were retained for detailed analysis.

The screening process followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement (Page et al., 2021). The adapted PRISMA flow diagram in Fig. 1 documents the four stages of identification, screening, eligibility, and inclusion. A total of 1639 records were initially identified, comprising 1518 from the four primary databases and 121 via manual forward and backward citation tracking. After removing 290 duplicates from database records, 1228 unique database records were subjected to title and abstract screening and 121 citation records were subjected to title and abstract screening. Of these, 1041 database records and 99 from citation tracking were excluded for failing to meet core inclusion criteria. The remaining 209 reports (187 database; 22 citation tracking) were sought for retrieval and assessed for eligibility at the full-text stage. At this phase, 149 reports were excluded (134 database; 15 citation tracking) for explicit reasons: insufficient empirical or conceptual depth (n = 71), a primary focus on patient-facing rather than healthcare-worker-facing technology (n = 38), an absence of clear linkage to resilience or continuance constructs (n = 27), or inaccessible full text (n = 13). This dual-stream selection process resulted in a final consolidated corpus of 60 studies retained for detailed analysis, consisting of 53 articles sourced from primary databases and 7 recovered through citation tracking.

Quality appraisal of included studies

To ensure that the synthesis was built on a credible evidentiary base, all 60 included studies were subjected to a structured quality appraisal. For empirical studies, the Mixed Methods Appraisal Tool (MMAT) version 2018 was applied (Hong et al., 2018), which provides validated criteria for quantitative, qualitative, and mixed-methods designs. For conceptual papers and prior reviews, the SANRA (Scale for the Assessment of Narrative Review Articles) criteria were used (Baethge et al., 2019). Each study was independently scored by two reviewers across the relevant criteria; only studies that met at least four of the five MMAT criteria (or three of the six SANRA criteria for reviews) were retained. All 60 eligible studies subjected to the appraisal successfully met these respective thresholds, resulting in zero exclusions due to quality scores. In addition, the outlet quality of each source was classified into three tiers based on indexing in Scopus/Web of Science, journal impact factor, and disciplinary recognition: Tier 1 (leading journals such as MIS Quarterly, Information Systems Research, JAMIA, Technovation, JMIR, BMC and Frontiers outlets, accounting for approximately 65% of the corpus), Tier 2 (specialised but reputable journals and major IS conferences such as ECIS, PACIS, AMCIS, accounting for approximately 25% of the corpus), and Tier 3 (emerging or specialised regional outlets, accounting for approximately 10% of the corpus). Tier 3 studies were retained only where they provided unique empirical evidence from under-represented contexts (e.g., low-resource healthcare settings in sub-Saharan Africa); their findings are reported with appropriate caveats. This tiered approach was used strictly to provide descriptive context regarding the structural distribution of the publications across the multidisciplinary field.

Analysis of selected literature

The analysis followed grounded theory coding principles to extract recurring concepts and relationships across studies. Each article was read thoroughly, and open coding was applied to identify key ideas. These open codes were then grouped into axial codes based on conceptual similarity. Finally, through iterative mapping and integration, the axial codes were refined into selective codes representing the main themes of digital resilience and continuous HIS use research. Alongside thematic coding, each study was categorized according to its research methodology (quantitative, qualitative, mixed, or conceptual), research method (e.g., case study, survey, interview, or simulation), and theoretical foundation. A master classification table was created to record notes, excerpts, and coding categories for each article.

To enhance the transparency of the grounded-theory coding process, an audit trail was maintained throughout the analysis. Two reviewers

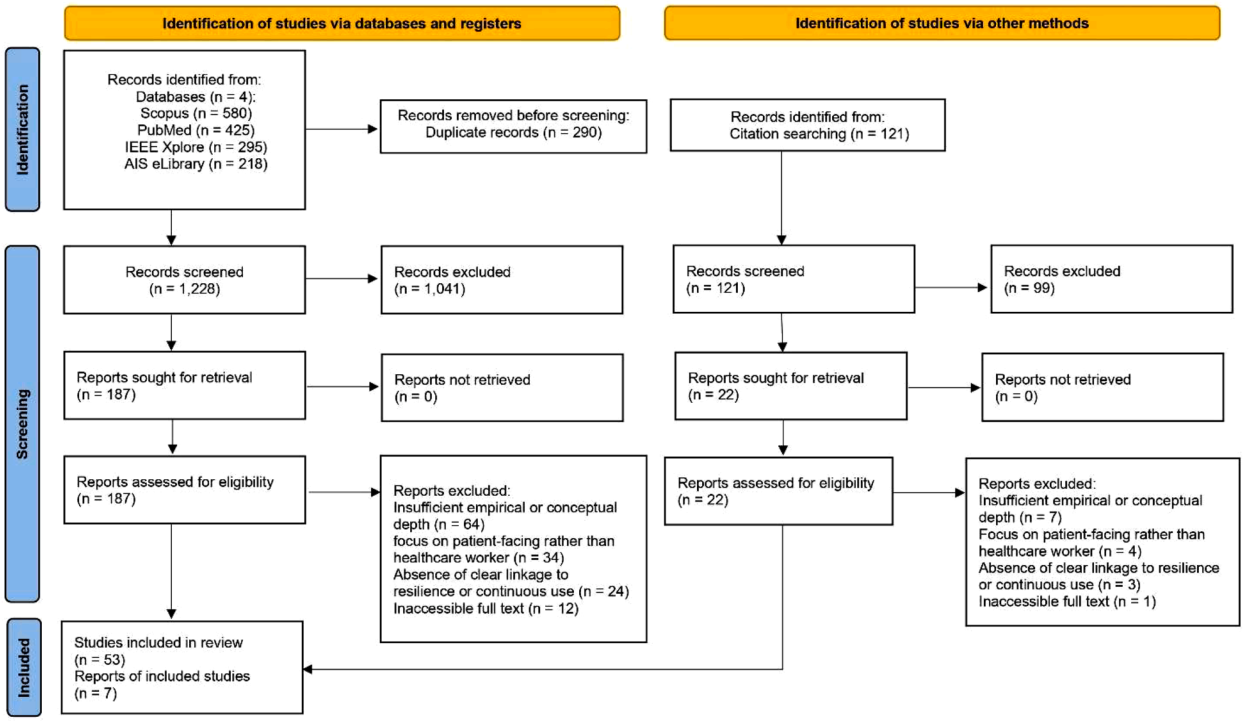

Fig. 1. Adapted PRISMA 2020 flow diagram of the study selection process.

independently performed open coding on the first 20 studies (one-third of the corpus); the resulting code lists were then compared, with an initial agreement of 83.6% on identified open codes. Discrepancies were discussed and reconciled into a consolidated coding scheme, which was subsequently applied to the remaining 40 studies by a primary reviewer and fully cross-checked and verified by the second reviewer to ensure absolute consistency.

Inter-rater reliability for the final classification of studies into themes was assessed using Cohen's $\kappa = 0.81$ (almost perfect agreement). An illustrative excerpt of the coding chain is presented here: the empirical finding that nurses resorting to personal mobile devices and unofficial applications to document and transmit patient data due to formal EHR system limitations (Abane et al., 2021; Mariwah et al., 2022) was mapped to the (open code: "shadow IT and personal device use") and grouped with similar open codes such as "paper persistence," "password sharing," and "personal smartphone documentation" into the axial code "Problem-Focused Coping and Workarounds," which in turn was integrated into the selective code (Theme 3) "Coping Mechanisms and Resilient Behaviors." All open codes, axial codes, supporting quotations, and the reviewers' reconciliation notes are documented in a supplementary audit-trail file available from the corresponding author on request.

This approach ensured that meanings and linkages between studies were systematically extracted and organized. The resulting thematic structure and methodological trends are presented in the next section, which summarizes the key findings of the review.

Results

Overview of healthcare worker digital resilience and continuous use of HIS research

This subsection outlines the distribution of the 60 reviewed publications across various journals (85.00%) and conference proceedings (15.00%), revealing a highly fragmented and multidisciplinary publication landscape. Among the individual outlets, PLoS ONE and the JMIR Journals group accounted for the highest representations at 5.00% each, followed closely by a robust cluster of specialized medical, nursing, and core Information Systems outlets—including BMC Nursing, BMC Health Services Research, Frontiers in Digital Health, Information Systems Journal, and Information Systems Research—each contributing 3.33% of the corpus. In terms of conference venues, the European Conference on Information Systems (ECIS) and the Pacific Asia Conference on Information Systems (PACIS) led with 3.33% each. This extensive dispersion across unique clinical, medical informatics, public health, and management platforms underscores the intersectional nature of digital resilience research in healthcare. However, this broad cross-disciplinary engagement also highlights the absence of a centralized publication hub, signaling a clear need for dedicated special issues or targeted conferences to better consolidate and advance scholarship in this domain. Table 1 provides a condensed summary of the top publication channels where multiple articles were recorded, while the complete, disaggregated list is available in Supplementary Table S4.

In terms of year of publication, the findings show mixed results.

Table 1
Summary of primary publication outlets (Outlets with $\geq$ 2 papers).

Journal / Conference name	Count	Percentage
Journals	51	85.00%
JMIR Journals (JMIR, Formative Res, Mental Health)	3	5.00%
PLoS ONE	3	5.00%
Health Economics and Management Review	2	3.33%
Information Systems Journal	2	3.33%
Information Systems Research	2	3.33%
Social Science & Medicine	2	3.33%
BMC Nursing	2	3.33%
BMC Health Services Research	2	3.33%
Frontiers in Digital Health	2	3.33%
Other Journals (31 unique outlets, 1 paper each)	31	51.67%
Conferences	9	15.00%
European Conference on Information Systems (ECIS)	2	3.33%
Pacific Asia Conference on Information Systems (PACIS)	2	3.33%
Other Conferences (5 unique venues, 1 paper each)	5	8.33%
Total	60	100.00%

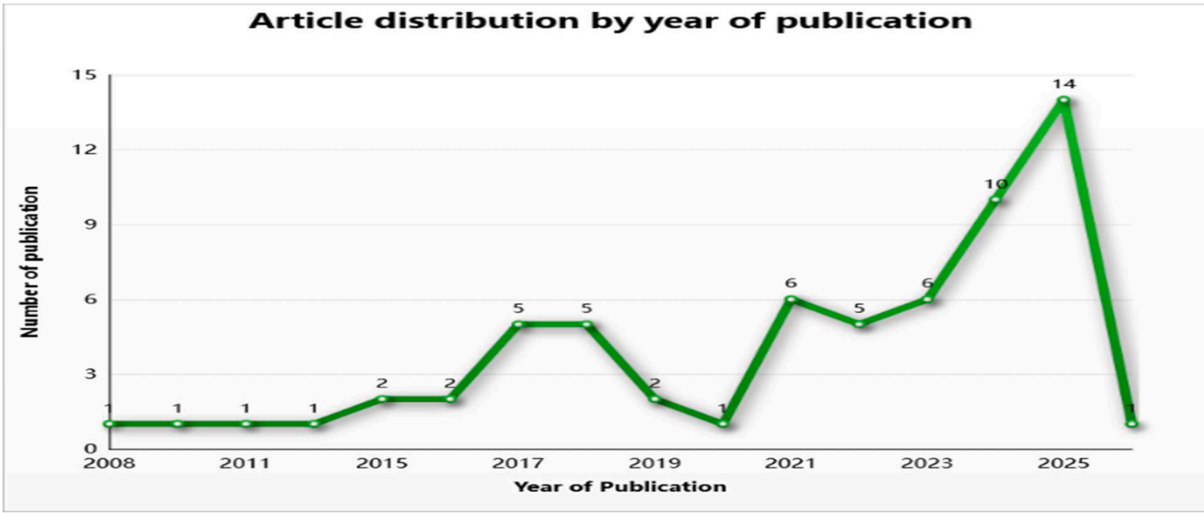

Fig. 2. Article distribution by year of publication (2008 – 2026).

Fig. 2 shows the distribution of articles from the year 2008 to 2026. The years 2025 and 2024 recorded the highest number of publications represented by 14 and 10 articles respectively. The years 2023 and 2021 recorded 6 articles each, while the years 2022, 2018, and 2017 recorded 5 articles each. Additionally, the years 2019, 2016, and 2015 recorded 2 articles each. Finally, the years 2026, 2020, 2014, 2011, 2010, and 2008 recorded 1 article each. While the chart displays a sharp downward trajectory for the year 2026, this artifact must be interpreted cautiously as a reflection of partial data collection bounded by our January 2026 search execution window.

The line chart shows publication frequency per year, with a notable surge from 2021 onward, peaking at 14 articles in 2025 and 1 article in 2026 because last search was in January 2026.

Themes in healthcare worker DR and HIS continuous use

This subsection addresses the study's research question 1 (RQ1). As presented in Table 2, research on healthcare worker digital resilience and continuous use of HIS can be classified into 4 main themes (see selective codes), namely *antecedents and enablers of digital resilience and HIS use, digital threats and technostress creators, coping mechanisms and resilient behaviors, and outcomes of digital resilience and continuous HIS use*. The antecedents and enablers of digital resilience and HIS use theme consists of 12 open and 3 axial codes while the digital threats and technostress creators theme contains 10 open and 3 axial codes. The coping mechanisms and resilient behavior's theme has 10 open and 3 axial codes. Lastly, the outcomes of digital resilience and continuous HIS use theme consists of 9 open and 3 axial codes. Each theme identified is further discussed as follows.

Table 2
Healthcare worker DR and HIS continuous use research themes.

Main Theme (Selective Code)	Axial Codes	Supporting Studies (Count)*	% of Corpus (N = 60)	Open Codes
1. Antecedents and Enablers of Digital Resilience and HIS Use	1.1 Human and Individual Factors 1.2 Technological and System Characteristics 1.3 Organizational and Environmental Support	48	80%	Big Five personality traits, digital literacy, self-efficacy, Intrinsic motivation. System usability, interoperability, interface complexity, ease of use. Top management support, IT assistance, peer training, funding.
2. Digital Threats and Technostress Creators	2.1 Cognitive and Workload Pressures 2.2 Workflow Disruptions and Misfits 2.3 Security, Privacy, and Ethical Concerns	44	73.33%	Techno-overload, multitasking, administrative burden, time pressure. System crashes, cumbersome data entry, poor connectivity. Data privacy risks, cyber threats, strict documentation rules.
3. Coping Mechanisms and Resilient Behaviors	3.1 Problem-Focused Coping and Workarounds 3.2 Emotion-Focused Coping 3.3 Dynamic Resilience Capabilities	41	68.33%	Risky Workarounds Shadow IT and personal device use, password sharing, Adaptive Coping proactive planning. Muting notifications, establishing work-life boundaries, peer venting. Anticipation, quick response teams, horizontal collaboration, flexibility.
4. Outcomes of Digital Resilience and Continuous HIS Use	4.1 Individual and Professional Outcomes 4.2 Patient and Care Outcomes 4.3 System-Level Transformation	52	86.67%	Continuance intention, job satisfaction, burnout reduction. Patient safety, reduced errors, continuity of care. Scaling care models, sustained feedback loops, organizational agility.

[*] Counts do not sum to 60 (and percentages do not sum to 100%) because primary studies routinely address multiple interlocking themes within a single paper.

Theme 1: antecedents and enablers of digital resilience and HIS use

This theme groups the foundational conditions enabling healthcare workers to cope with digital disruptions and sustain HIS use. Drawing on socio-technical systems theory (Bostrom and Heinen, 1977) continuous use depends on aligning human capabilities, technological attributes, and organizational support; the axial codes are human and individual factors, technological and system characteristics, and organisational and environmental support. Rather than evaluating these antecedents uniformly, they are critically synthesized across distinct analytical layers based on their empirical weight and operational boundaries to evaluate their baseline certainty and context-dependence.

At the macro-structural and meso-relational levels, the literature supports basic technical infrastructure and relational organizational support as prerequisites for digital resilience. Technical infrastructure such as reliable electricity, robust hardware, and continuous network connectivity is a requirement (Doctor et al., 2025; Okronipa and Nyame, 2024; Simbini et al., 2026); erratic power and network blackouts instantly paralyze digital workflows, causing delayed care and forcing a reversion to paper tracking (Mensah et al., 2023). Similarly, meso-level facilitating conditions, including continuous practical end-user training, vertical top-management commitment, and highly responsive IT helpdesk assistance, directly buffer user anxiety and prevent system abandonment when glitches occur during high-stress clinical shifts (Alsyouf et al., 2024; Golz et al., 2021; Maraj et al., 2024). Beyond formal organizational channels, relational factors provide a vital grassroots safety net; horizontal peer support networks, collective mentoring, and shared tacit knowledge allow clinical staff to bridge training gaps with situation-specific solutions, reinforce collective competence, and buffer the immediate strains of technostress (Golz et al., 2021; Li et al., 2025).

At the technical level, specialized system attributes are heavily validated as critical determinants of continuous use, though their operational impact remains deeply context-dependent (Kutney-Lee et al., 2019; Maraj et al., 2024). Drawing on the Technology Acceptance Model (TAM) (Davis, 1989), Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003), and the DeLone and McLean IS Success Model (Delone and McLean, 2003), attributes such as system usability, information quality, performance expectancy, and effort expectancy critically shape continuance intentions. Intuitive user interfaces, clear data navigation, high data accuracy, and transaction stability reduce adoption friction and administrative drudgery (Kutney-Lee et al., 2019). Furthermore, recent empirical evidence highlights that technological reliability and interface transparency are foundational drivers of digital trust and system credibility among frontline clinicians, establishing the baseline psychological security required for sustained technology engagement (Phan, Le, et al., 2026, 2026). Conversely, overly complex workflows, unreliable connectivity, or rigid data configurations trigger systemic rejection and drive staff to seek alternative workarounds or abandon the platform entirely (Abane et al., 2021; Kutney-Lee et al., 2019).

Finally, micro-level human and psychological traits represent an emergent, more speculative line of inquiry whose broad generalizability across diverse clinical cohorts is still under empirical validation (Alsyouf et al., 2024; Li et al., 2025; Mo et al., 2025). Applying the Five-Factor Model (FFM) (Costa and McCrae, 1992), individual traits such as openness, extraversion, and conscientiousness positively influence technology acceptance, lower learning curves, and ease system mastery, whereas neuroticism systematically amplifies technostress and undermines facilitating conditions (Alsyouf et al., 2024; Ye et al., 2024). Frontline professionals possessing robust adaptive self-efficacy interpret unexpected system updates, technological disruptions, or strict compliance rules as manageable professional challenges; this intrinsic psychological fortitude motivates them to explore, adapt, and problem-solve rather than experience emotional exhaustion, data errors, or total system abandonment (Alsyouf et al., 2024; Li et al., 2025; Mo et al., 2025).

Theme 2: digital threats and technostress creators

This theme synthesizes the literature examining the negative, unintended consequences and systemic vulnerabilities introduced by the widespread adoption of digital health technologies. While health information systems (HIS) are designed to streamline care delivery and optimize patient outcomes (Okronipa and Nyame, 2024), their implementation creates psychological, operational, and technological hazards that act as significant barriers to continuous system use (Guo et al., 2018; Ragu-Nathan et al., 2008; Tarafdar et al., 2015; Zheng et al., 2016). These hazards can be systematically evaluated across three core sub-themes: cognitive and workload pressures, workflow disruptions and sociotechnical misfits, and security, privacy, and ethical concerns.

Regarding cognitive and workload pressures, studies highlight the psychological and cognitive strain healthcare workers experience when adapting to continuous digital demands, a phenomenon conceptualized as "technostress" (Ragu-Nathan et al., 2008; Tarafdar et al., 2015). This construct is frequently operationalized across five dimensions: techno-overload, techno-invasion, techno-complexity, techno-uncertainty, and techno-insecurity (Tarafdar et al., 2015). Techno-overload occurs when the volume and pace of digital data force healthcare professionals to work faster and longer (Tarafdar et al., 2015). Observational studies utilizing tools such as the NASA Task Load Index (NASA-TLX) demonstrate that interruptions during electronic health record (EHR) tasks lead to rapid task-switching, which decays the clinician's memory of the initial task, prolongs response times, and overwhelms cognitive capacities (Kim and Lee, 2021; Shan et al., 2023). Alongside overload, techno-invasion describes how constant connectivity blurs the boundaries between professional and personal life, exposing workers to endless tasks and push notifications outside scheduled shifts (Shan et al., 2023; Wallace, 2017). Techno-complexity arises from non-intuitive medical IT systems that intimidate users with complex jargon and convoluted navigation, requiring significant cognitive effort to learn new workflows (Wirkkala et al., 2024). Compounding this are techno-uncertainty, driven by rapid software upgrades that prevent the development of a stable base of experience, and techno-insecurity, relating to the fear of professional obsolescence (Kim and Lee, 2021). The cumulative effect of these cognitive pressures is directly linked to elevated rates of job dissatisfaction, emotional exhaustion, and provider burnout (Wallace, 2017).

Transitioning to workflow disruptions and sociotechnical misfits, the literature examines the operational friction that occurs when rigid digital designs clash with the dynamic realities of clinical practice. Sociotechnical theory emphasizes that performance depends on alignment between technology, medical staff, and the care context (Bostrom and Heinen, 1977; Sittig and Singh, 2010); however, a pervasive misfit is usually observed (Beerepoot and Weerd, 2018; Kutney-Lee et al., 2019). Standardized checklists or computerized physician order entry (CPOE) systems are often poorly tailored to specific hospital wards, disrupting natural clinical workflows (Agarwal et al., 2010). This friction is exacerbated by a lack of system integration and interoperability across laboratory information systems (LIS), radiology information systems (RIS), and core EHR platforms (Provenzano et al., 2024; Tzitzili et al., 2023). This fragmentation forces medical staff to perform duplicate documentation, which wastes clinical time, increases administrative burden, elevates medical error risks, and compromises the continuity of care (Dehnavieh et al., 2019). In low-resource settings and emerging economies, these workflow disruptions are compounded by infrastructural deficits, such as unstable electricity and intermittent network connectivity (Heeks and Ospina, 2019; Mensah et al., 2023). This doubles the workload upon system restoration due to retrospective data entry, introducing opportunities for transcription errors (Argaw et al., 2020). Additionally, a lack of hardware creates physical bottlenecks on the ward, forcing staff to queue for workstation access and delaying patient care (Simbini et al., 2026).

The final domain encompasses security, privacy, and ethical concerns. As healthcare organizations digitize repositories of sensitive patient data, they become frequent targets for cyber threats, such as ransomware, phishing, and supply chain attacks (Argaw et al., 2020). The theft of personal medical data is a vulnerability exacerbated by reliance on legacy software and interconnected Internet of Medical Things (IoMT) devices that lack robust security configurations (Garcia-Perez et al., 2023). System downtime caused by a cyberattack can lead to an operational standstill of hospital operations, paralyzing digital solutions and endangering patient safety (Argaw et al., 2020). Consequently, reliance on workarounds—such as using shadow IT or sharing passwords to bypass slow login processes introduces internal privacy risks (Abane et al., 2021; Mariwah et al., 2022).

Across the reviewed studies on this theme, technostress creators act as the primary behavioral pivot determining whether an individual pursues continuous HIS use, executes workarounds, or abandons the platform. When techno-overload and complexity breach cognitive thresholds, they directly drive behavioral discontinuance intentions or total system rejection (Guo et al., 2018). To maintain care continuity despite these system frictions, clinicians routinely resort to risky workarounds like password sharing or shadow IT (Abane et al., 2021; Mariwah et al., 2022); conversely, when adequately supported by organizational resources, these same threats can trigger dynamic resilience mechanisms through collaborative peer troubleshooting networks (Li et al., 2025; Mo et al., 2025).

Theme 3: coping mechanisms and resilient behaviors

The coping mechanisms and resilient behaviors theme groups studies that investigate how healthcare professionals actively manage, navigate, and psychologically process the continuous disruptions and stressors introduced by Health Information Systems (HIS). Rather than passively absorbing the technological burdens of the digital era, healthcare workers deploy a complex spectrum of strategies to maintain patient safety, ensure operational efficiency, and preserve their own mental well-being. The literature consistently frames these behaviors as continuous cognitive and behavioral efforts used to manage internal and external demands that are perceived as exceeding an individual's resources (Beh and Loo, 2012; Mariwah et al., 2022). Within this sociotechnical ecosystem, coping strategies are broadly categorized into problem-focused mechanisms, emotion-focused mechanisms, and dynamic, system-level resilience capabilities (Abane et al., 2021; Beh and Loo, 2012).

Problem-focused coping involves active, instrumental measures taken by healthcare workers to alter the stressful situation, eliminate the source of the stress, or solve the immediate technological problem (De Vries et al., 2025; Feng et al., 2024). In the highly demanding and time-critical environment of healthcare, this often manifests as proactive work strategies designed to streamline daily interactions with digital systems. For instance, workers frequently engage in anticipatory planning, such as logging into multiple platforms (e.g., electronic health records, communication channels, and clinical guidelines) early in the morning before their first patient arrives, or creating personalized templates and shortcuts to expedite mandatory data entry (Golz et al., 2021).

When formal problem-focused strategies fail—often because the HIS is poorly integrated, excessively complex, or fundamentally misaligned with clinical workflows—healthcare workers frequently resort to workarounds. Workarounds are defined as goal-driven adaptations, improvisations, or alternative procedures employed by users to bypass obstacles and achieve desired outcomes when there is a severe misfit between the prescribed computer-based processes and the realities of frontline medical work (Beerepoot and Weerd, 2018; Oprea et al., 2025; Rubbio et al., 2019). In essence, when a designed digital pathway is blocked or overly cumbersome, a workaround provides an alternative route to the same clinical goal without completely removing the underlying systemic block (Doctor et al., 2025). These deviations from standard operating procedures are incredibly common in hospital settings and represent a highly localized, problem-solving response to sociotechnical friction (Beerepoot and Weerd, 2018).

The literature provides numerous examples of these behavioral adaptations, highlighting their pervasive and multifaceted nature. Healthcare workers frequently exhibit "paper persistence," opting to manually write down patient vitals and clinical notes on paper during ward rounds rather than navigating a cumbersome Computer on Wheels (COW) or a slow-loading mobile interface (Beerepoot and Weerd, 2018). In other instances, professionals construct unauthorized "shadow IT" systems, such as custom using apps on their mobile phones, to collaboratively track patient statuses and workflows because the official HIS lacks the necessary functionalities for shared editing or mid-process transparency (Argaw et al., 2020). Additionally, to circumvent slow authentication processes or the lack of proper role-based access, staff members frequently engage in the illicit sharing of passwords or logging in under a colleague's account (Doctor et al., 2025).

Workarounds embody a profound paradox within the realm of digital resilience. On one hand, they are viewed positively as a testament to the ingenuity and dedication of healthcare workers, ensuring that immediate, life-saving patient care is not halted by software glitches or poorly designed interfaces (Beerepoot and Weerd, 2018). They act as a vital feedback mechanism, illuminating the exact points where a system fails its users and providing a grassroots blueprint for future system redesign (Beerepoot and Weerd, 2018). On the other hand, these problem-focused adaptations carry significant systemic risks. By circumventing official digital protocols, workers inadvertently introduce severe vulnerabilities regarding data security, patient privacy, and clinical standardization, potentially leading to medication errors, fragmented patient records, and compromised information governance (Doctor et al., 2025).

When problem-focused coping is either impossible or insufficient—such as when facing mandatory system updates, persistent network outages, or the relentless influx of digital notifications—healthcare workers shift heavily toward emotion-focused coping. This strategy aims to regulate, minimize, or process the psychological and emotional distress associated with technostress, rather than attempting to alter the unchangeable technological environment (Kräft et al., 2024). Because healthcare professionals cannot simply abandon the mandated HIS, they must find ways to internally buffer the anxiety, frustration, and cognitive fatigue that these systems frequently induce (Kräft et al., 2024).

A primary emotion-focused strategy utilized by medical staff is psychological distancing and strict boundary management. To combat "techno-invasion"—the blurring of professional and personal lives facilitated by mobile health technologies—workers deliberately limit their exposure to digital tools outside of their scheduled shifts (Kräft et al., 2024). Strategies include aggressively muting work-related notifications, refusing to install hospital communication applications on personal smartphones, and completely segregating their personal and professional digital domains to protect their recovery time (Kräft et al., 2024). While psychological avoidance can sometimes be maladaptive if used to permanently ignore critical problems, in the context of relentless digital connectivity, this deliberate technological disengagement acts as a necessary self-preservation tactic to prevent total emotional exhaustion and burnout (Tahara et al., 2020).

Social and peer support serves as another indispensable emotion-focused coping mechanism. In high-pressure medical environments where formal IT support is often perceived as inadequate, delayed, or disconnected from clinical realities, the informal networks among nurses and physicians become a critical lifeline (Golz et al., 2021). Sharing frustrations, venting about system absurdities, and using dark humor with colleagues who understand the exact nuances of the technological struggle provide a profound sense of validation and community. This horizontal, peer-to-peer solidarity not only facilitates rapid, informal knowledge sharing to bridge training gaps, but it also significantly diminishes the isolating effects of technostress, reinforcing the collective psychological resilience of the ward (Rubbio et al., 2019).

To systematically cultivate these emotional and cognitive defenses, numerous studies investigate the efficacy of formal psychological interventions and digital resilience training programs. Healthcare organizations are increasingly deploying digital platforms, smartphone applications, and blended learning modules that utilize evidence-based psychological frameworks such as Cognitive Behavioral Therapy (CBT), Acceptance and Commitment Therapy (ACT), and Mindfulness-Based Resilience Training (MBRT) (Ang et al., 2022; Larsson et al., 2025). These programs train workers in cognitive flexibility, teaching them how to reframe negative technological experiences, regulate their physiological stress responses through breathing and meditation, and maintain an optimistic, realistic mindset (Ang et al., 2022). Specific mHealth applications provide tools ranging from positive affect journaling and gratitude exercises (e.g., the "Three Good Things" intervention) to real-time burnout assessments, equipping workers with accessible, on-demand resources to actively manage their mental health (Larsson et al., 2025).

These individual coping strategies and formal interventions ultimately aim to build Psychological Capital (PsyCap), a composite construct comprising hope, resilience, optimism, and self-efficacy (Burns et al., 2017). In the context of digital health, adaptive computer self-efficacy—the deeply held belief in one's own capability to successfully navigate, master, and troubleshoot digital tools—acts as a fundamental driver of resilient behavior (Burns et al., 2017; Mo et al., 2025). When healthcare workers possess robust PsyCap, they are significantly more motivated to explore new platforms, endure the initial friction of steep learning curves, and interpret digital disruptions as manageable professional challenges rather than insurmountable, career-ending threats (Burns et al., 2017; Gröschke et al., 2022). This intrinsic psychological fortitude allows them to transition from a state of mere survival and technological resistance to a state of active, positive adaptation.

Beyond individual psychological traits and coping behaviors, the literature emphasizes that resilient behaviors must be conceptualized as dynamic capabilities at the team and organizational levels. Resilience engineering posits that the safety and continuous performance of a healthcare system rely on four interrelated potentials: the ability to anticipate future disruptions, monitor current systemic threats, respond adaptively to ongoing crises, and learn from past successes and failures (Ambrose et al., 2024). From this macroscopic perspective, resilience is not just an individual clinician's ability to "bounce back" from a stressful shift, but rather the collective capacity of the entire healthcare team to function as a cohesive, interconnected, and conscious entity capable of collective problem-solving (Ambrose et al., 2024; Gilson et al., 2020).

Team cohesion, altruistic behaviors, and multi-skilled workforce flexibility are the bedrock of this dynamic resilience. During periods of extreme systemic shock—such as the unprecedented operational strains induced by the COVID-19 pandemic—healthcare professionals frequently step outside their rigidly defined professional boundaries to assist one another (Ambrose et al., 2024; Larsson et al., 2025). This horizontal collaboration ensures that when a digital infrastructure fails, a network goes offline, or a critical clinical pathway is bottlenecked by software, the team can seamlessly redistribute the workload. They rely on their shared tacit knowledge, profound clinical expertise, and mutual trust to maintain patient care, effectively absorbing the shock that the technological failure introduced (Ambrose et al., 2024; Rubbio et al., 2019).

Ultimately, the continuous deployment of these problem-focused workarounds, emotion-focused boundary settings, and dynamic team collaborations leads to a state of system-level transformation. Resilient healthcare systems do not simply endure a digital disruption and revert to a stagnant baseline; rather, they engage in a process of continuous learning, unlearning old rigid behaviors, and fundamentally adapting their structural realities (Boh et al., 2023; Gilson et al., 2020). By recognizing informal coping mechanisms as critical diagnostic data rather than mere policy violations, and by institutionalizing the peer support networks that workers naturally forge, organizations can proactively redesign both their technologies and their care models (Beerepoot and Weerd, 2018). Crucially, workarounds must not be viewed simplistically as unmitigated signs of individual resilience; rather, they exist in deep tension as acute signals of systemic IT infrastructure failure and structural misfits. When clinicians routinely deploy unauthorized bypasses, these behaviors risk institutionalizing a "normalization of deviance" where substandard operational shortcuts become standard operating procedures, systematically masking underlying software defects from corporate IT oversight. Furthermore, these adaptive shortcuts frequently trigger severe data privacy and regulatory compliance violations, effectively executing a hidden risk transfer where the legal and cognitive burden of navigating poorly configured, non-interoperable platforms is shifted entirely from the organization onto individual frontline workers.

This dual nature of workarounds establishes the empirical foundation for a non-linear, inverted-U relationship between frontline resilience behaviors and systemic HIS performance. While a moderate deployment of adaptive workarounds optimizes immediate patient care by bypasses rigid system rigidities, an escalation into excessive, uncoordinated deviations introduces severe data vulnerabilities that ultimately degrade system-wide integrity. Consequently, this paradox suggests that maximum digital resilience is achieved not through total system compliance or unmitigated workarounds, but at an optimal behavioral equilibrium point.

Theme 4: outcomes of digital resilience and continuous HIS use

The outcomes of digital resilience and continuous Health Information Systems (HIS) use theme groups studies that examine the ultimate consequences and long-term impacts of successful technological adaptation within healthcare environments. Rather than viewing the deployment of digital tools as a finite event, the literature consistently emphasizes that when healthcare professionals successfully navigate digital disruptions and sustain their use of health technologies, it generates profound, far-reaching benefits. These benefits ripple outward from the individual practitioner to the patient, and ultimately to the macroeconomic structure of the healthcare organization itself. Consequently, the axial codes under this theme include individual and professional outcomes, patient and care outcomes, and system-level transformation.

Studies within the individual and professional outcome’s sub-theme highlight the profound psychological, operational, and behavioral shifts that occur when healthcare workers successfully develop digital resilience. A primary outcome at this level is the establishment of "continuance intention," which represents the user’s conscious decision to persist with a technology long after its initial implementation (Alsyouf et al., 2024). In the healthcare sector, initial acceptance of a system like an Electronic Health Record (EHR) does not guarantee sustained engagement; without resilience, systems often face the "shelf-ware" problem, characterized by underutilization, resistance, or total rejection (Bagherian and Sattari, 2022; Jamaluddin and Abawajy, 2021). However, when nurses and physicians are equipped with adequate organizational support and possess the necessary personal resilience traits, they transition from a state of resistance to one of committed, continuous use(Alsyouf et al., 2024). This sustained engagement is deeply intertwined with enhanced professional empowerment and the development of "digital thinking." Nurses with a high level of digital resilience do not merely survive the digital environment; they leverage digital skills and information literacy to actively seek solutions to nursing problems, utilizing digital principles as guiding views for clinical work (Mo et al., 2025; Rahman and Islam, 2024). This intrinsic empowerment allows them to confidently navigate complex digital risks, bridging the gap between basic digital literacy and advanced technological mastery (Mo et al., 2025; Sun et al., 2022).

Furthermore, the successful development of digital resilience leads to highly favorable occupational health outcomes, most notably the mitigation of technostress and the reduction of burnout. The relentless pressure of digital environments frequently leads to emotional exhaustion, depersonalization, and a reduced sense of professional accomplishment (Larsson et al., 2025). However, empirical evidence demonstrates that formal digital resilience interventions—including web-based cognitive behavioral therapy, mindfulness programs, and self-directed e-learning—significantly enhance the psychological well-being of healthcare personnel (Larsson et al., 2025; Sun et al., 2022). These interventions foster psychological flexibility and perseverance, enabling workers to establish strict work-life boundaries and reduce emotional distress (Li et al., 2025). Operationally, this improved individual resilience fundamentally alters the relationship between the worker and their workload. While the introduction of DHTs often initially increases workload due to steep learning curves, resilient healthcare workers who continuously use the systems eventually experience a significant reduction in administrative burdens (Jeilani and Hussein, 2025). By mastering the technology, professionals streamline their daily tasks, optimize their time management, and reduce the cognitive friction associated with digital navigation (Jeilani and Hussein, 2025; Mensah et al., 2023). In turn, this heightened performance efficiency mediates the stress of high-pressure clinical environments, leading to greater job satisfaction, increased organizational commitment, and a significantly lower intention to leave the profession (Provenzano et al., 2024).

Transitioning from the provider to the recipient of care, the patient and care outcomes sub-theme examines how the resilient, continuous use of HIS directly elevates the quality, safety, and efficiency of medical service delivery. When healthcare workers effectively integrate digital tools into their routines without succumbing to workflow disruptions, the most immediate beneficiary is the patient. The literature heavily underscores improved patient safety and the dramatic reduction of medical errors as paramount outcomes of sustained HIS utilization (Woldemariam and Jimma, 2023). In a resilient digital ecosystem, the structural advantages of electronic documentation actively protect patients from human error; for instance, the elimination of illegible handwriting and manual transcription issues directly curtails medication administration mistakes (Mensah et al., 2023). Furthermore, systems that provide clinical decision support, automated alerts, and real-time tracking of vital signs act as a critical safety net, prompting clinicians to adhere to standardized care protocols and preventing fatal oversights during high-stress encounters (Simbini et al., 2026).

The continuous, proficient use of digital health technologies also fundamentally enhances the continuity and quality of care. In fragmented medical environments, patients frequently transition between different wards, specialists, and facilities, which traditionally poses a severe risk of information loss. However, when health professionals consistently utilize interoperable EHRs and mobile health platforms, comprehensive patient histories become instantly accessible across the entire healthcare continuum (Simbini et al., 2026). This seamless information sharing fosters robust interprofessional communication, breaking down traditional clinical silos and ensuring that any incoming clinician can review a patient's previous treatments, laboratory results, and diagnoses without delay (Mensah et al., 2023; Provenzano et al., 2024). Consequently, this rapid and easy access to real-time data drastically accelerates clinical decision-making. Physicians and nurses report that the ability to retrieve accurate patient profiles with a single click shortens the consultation process, reduces patient waiting times, and eliminates redundant diagnostic testing (Mensah et al., 2023). By liberating clinicians from the administrative drudgery of searching for lost paper files, digital resilience ensures that medical staff can redirect their focus and time toward direct, human-centered patient interaction, thereby preserving the vital "human touch" in a highly digitized era (Mo et al., 2025). Furthermore, the continuous and secure use of these systems enforces patient privacy and data confidentiality, fostering a deeper sense of trust between the patient and the healthcare institution (Mensah et al., 2023).

The final sub-theme, system-level transformation, explores the macroscopic consequences of digital resilience, illustrating how local adaptations and continuous technology use aggregate to fundamentally redesign the broader healthcare ecosystem. The literature emphasizes that a truly resilient health system does not merely "bounce back" to its pre-disruption state after a crisis or technological shock; rather, it "bounces forward," utilizing the disruption as a catalyst for long-lasting systemic improvement and transformation (Pilosof et al., 2025). One of the most significant indicators of this transformation is the organization's ability to scale new models of care. For example, during the extreme operational stresses of the COVID-19 pandemic, digitally resilient healthcare communities rapidly experimented with, refined, and permanently integrated novel service delivery methods, such as inpatient telemedicine, remote patient monitoring, and online healthcare communities (Pilosof et al., 2025). Professionals transcended traditional clinical roles to become digital innovators, successfully transferring patients between physical and digital channels and redesigning clinical pathways to manage massive surges in demand (Boh et al., 2023). This capacity to innovate at scale ensures that the healthcare system is not merely reactive, but proactively adaptive to emerging demographic pressures and future crises (Pilosof et al., 2025).

System-level transformation is heavily sustained by dynamic feedback loops and continuous learning. Digital resilience is conceptualized as an ongoing, cyclical process where healthcare workers continuously interact with technologies, experience friction, and provide vital feedback (Mo et al., 2025). In a resilient organization, this grassroots feedback is not ignored but is systematically absorbed by management and software developers to optimize system interfaces, fix interoperability gaps, and redesign workflows (Mo et al., 2025). This sustained dynamic feedback ensures that the digital infrastructure constantly evolves in alignment with actual clinical realities, turning static software into an agile, highly customized clinical asset (Beerepoot and Weerd, 2018). This culture of collective problem-solving and open innovation dissolves hierarchical rigidities, fostering a collaborative governance model where multidisciplinary teams jointly construct the future of their digital work environment (Pilosof et al., 2025).

Ultimately, the aggregate outcome of digital resilience is the creation of a secure, equitable, and highly adaptable healthcare ecosystem.

Research methodologies and methods in digital resilience and HIS use research

This subsection presents findings on methodologies and methods used in digital resilience and continuous HIS use research. The findings as presented in Table 3 address research question 2 (RQ2). We adapted Alavi and Carlson (1992) research strategy classification framework during this analysis since it has extensive coverage of research methodologies and methods. The findings show that a notable number of the

Table 3
Research Methodologies and methods in Digital Resilience and HIS use research.

Research Classification	Research Methodologies and Methods	Count	Percentage
Empirical			
Quantitative	Survey / Cross-sectional	20	33.33%
Qualitative	Interview	14	23.33%
	Case study	8	13.33%
Mixed-Methods	Survey & Interview / Focus Group	4	6.67%
Non-Empirical			
Conceptual / Review	Systematic, Scoping & Literature Reviews	14	23.33%
Total		**60**	**100%**

research articles are systematic reviews, scoping reviews, or conceptual in nature and therefore non-empirical (23.33%). While this indicates a strong effort to map and synthesize this emerging field, it also means a portion of the literature is not directly testing its conceptualizations to determine their applicability, efficacy, and performance in practice.

In terms of empirical studies, the survey (33.33%) and interview (23.33%) methods are the widely used approaches under the quantitative and qualitative methodologies. These approaches are followed by the case study (13.33%) and mixed-methods (6.67%) respectively. Under the quantitative spectrum, researchers predominantly utilize cross-sectional surveys analyzed via Structural Equation Modeling (SEM) or Partial Least Squares (PLS) to statistically test theoretical frameworks like UTAUT, the Technology Acceptance Model (TAM), and the DeLone and McLean IS Success Model. Under the qualitative spectrum, semi-structured interviews and case studies are heavily utilized, often drawing on Grounded Theory or Interpretative Phenomenological Analysis to deeply explore users' lived experiences. Mixed-methods approaches were also employed to triangulate findings, combining survey data with in-depth focus group discussions or interviews.

With regards to empirical studies, the result shows that digital resilience and continuous HIS use research is methodologically mixed, balancing qualitative explorations of lived experiences with a continued prominence of quantitative surveys. One probable reason for the use of qualitative methods like interviews and case studies is the need to understand complex, context-dependent human behaviors—such as technostress, emotional coping mechanisms, and sociotechnical workarounds—which are difficult to capture fully through quantitative surveys alone. Thus, the interview and case study methods seem to be the most suitable options to access primary data from frontline healthcare participants navigating actual HIS disruptions. In fact, most qualitative studies have called for quantitative testing of their propositions and longitudinal validations, which explains the parallel prominence of quantitative surveys designed to validate the exact relationships between system characteristics, resilience, and continuous use.

Beyond the methods represented in Table 3, the review identified a striking absence of certain research approaches in the literature on healthcare workers' digital resilience and continuous HIS use. Action research, ethnography, longitudinal designs, and system-log analysis were entirely absent from the corpus (n = 0 in each category). This absence is analytically meaningful because each of these approaches is well-suited to capturing the temporal, contextual, and behavioural dimensions of digital resilience that cross-sectional self-report cannot. Action research could enable researchers to co-design and iteratively refine resilience-building interventions in real clinical settings; ethnography could illuminate the embodied, situated practices through which healthcare workers improvise around digital constraints; longitudinal studies could trace how resilience develops, decays, or transforms over implementation cycles; and log-data analysis could provide objective traces of actual continuance behaviour beyond self-report. Their absence therefore constitutes a substantive methodological gap that we revisit in detail in the future research agenda section.

Theories used in healthcare worker DR and HIS continued use research

This subsection addresses research question 3 (RQ3) of this paper. With respect to theory, we examined the sampled papers to identify their theoretical underpinnings. The findings from the analysis as presented in Table 4 show that 46.67% of the publications did not use any theory. As such, we classified such articles under the "no theory" category. Among the articles that used theories, the Job Demands-Resources (JD-R) model recorded the highest usage of 10%, followed by the Technology Acceptance Model (TAM) and UTAUT framework (8.33%), the Transactional Theory of Stress and Coping (6.67%), and the Socio-Technical Systems (STS) theory (5.00%). The Dynamic Capabilities theory, Diffusion of Innovations (DOI) theory, and the Five-Factor Model

Table 4
Theories used in healthcare worker Digital Resilience and HIS continuous use research.

Theories	Count	Percentage
Job Demands-Resources (JD-R) model	6	10.00%
Technology Acceptance Model (TAM) & UTAUT	5	8.33%
Transactional Theory of Stress and Coping	4	6.67%
Socio-Technical Systems (STS) theory	3	5.00%
Dynamic Capabilities theory	2	3.33%
Diffusion of Innovations (DOI) theory	2	3.33%
Five-Factor Model (FFM)	2	3.33%
Theories that appeared only once (listed in text description)	8	13.33%
No theory	28	46.67%
Total	**60**	**100%**

(FFM) each recorded 3.33%. The rest of the theories such as Normalization Process Theory, Actor-Network theory, Protection Motivation Theory, Social Cognitive Theory, Theory of Planned Behavior, DeLone and McLean IS Success Model, Conservation of Resources (COR) theory, and Resilience Engineering framework each recorded appeared once representing 13.33%.

The revelations from this analysis are that: (1) nearly half of the digital resilience and continuous HIS use studies do not use theories and (2) this research domain lacks its own native theories. These revelations could be attributed to the relative newness of the digital resilience concept in the healthcare context. Even though healthcare worker digital resilience is a relatively new construct, it is still important for researchers to make the effort to build domain-specific theories for significant contributions to the development of the research area.

A closer cross-tabulation of theory use against research design reveals that the atheoretical tendency is not evenly distributed across methodological approaches. Of the 28 studies classified as "no theory," 19 are qualitative (interview, case study), 5 are systematic, scoping, or narrative reviews, 3 are mixed-methods, and only 1 is a cross-sectional survey. In other words, virtually all quantitative empirical studies in the corpus are theory-informed, typically anchored in TAM, UTAUT, JD-R, or the DeLone and McLean model, because variance-based hypothesis testing requires an a priori theoretical model. By contrast, the qualitative and review-based studies most often adopt an inductive or descriptive stance, presenting findings without explicit theoretical scaffolding. This pattern suggests that the theoretical deficit in the field is concentrated in interpretive and synthetic work rather than in quantitative inquiry. It also points to a missed opportunity: qualitative studies are precisely the works best placed to generate the domain-specific theory that the field currently lacks. Future qualitative research should therefore move beyond rich description to theory elaboration or grounded theorising about how healthcare workers' digital resilience is constituted and sustained.

Framework for healthcare worker DR and continuous HIS use

This section presents an integrative conceptual framework designed to transcend basic descriptive mapping by operationalizing the systemic, relational dynamics of digital resilience. Rather than serving as a static taxonomy of isolated variables, the framework represents an inductive, theory-building model that captures the causal and non-linear interactions across research themes, methodologies, and theories. As illustrated in Fig. 3, the framework treats digital resilience as a dynamic socio-technical process where foundational antecedents condition threat exposure, threats trigger a complex spectrum of coping behaviors, and localized outcomes loop back to fundamentally reconfigure institutional capacities. By articulating these interdependencies through six testable, directional propositions, the framework transitions from a retrospective classification of past literature into a forward-looking, falsifiable foundation for domain-specific theory elaboration.

Before presenting the framework, we briefly examine the cross-

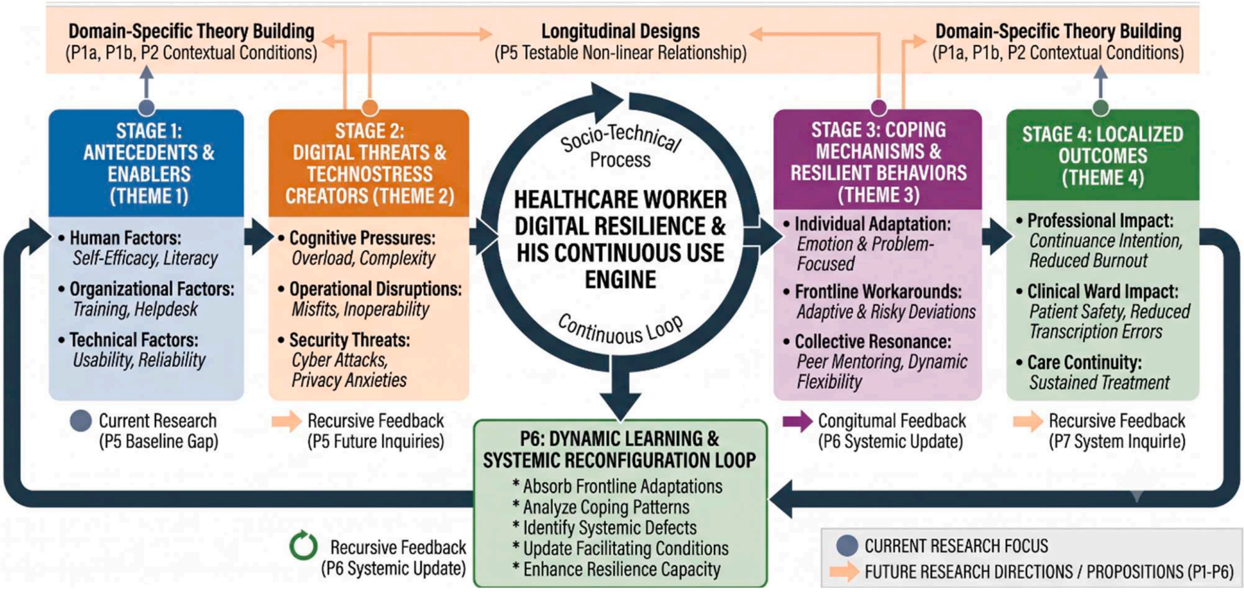

Fig. 3. Framework for Healthcare worker Digital Resilience and HIS continued use.

theme interrelationships that emerged from the synthesis, because the four themes do not operate as discrete categories. The synthesis surfaced four recurring patterns of interaction. First, antecedents (Theme 1) condition exposure to threats (Theme 2): healthcare workers with low digital literacy or weak organisational support experience the same techno-stressors more acutely than colleagues with high self-efficacy and strong IT support. Second, threats (Theme 2) trigger coping behaviours (Theme 3): persistent workflow misfits, infrastructural failures, and techno-overload drive workarounds and emotion-focused coping, which are themselves contingent on the human and organisational antecedents of Theme 1. Third, coping (Theme 3) shapes outcomes (Theme 4) in a dual direction: problem-focused workarounds preserve immediate continuity of care but introduce systemic risks (data fragmentation, privacy violations) that can undermine longer-term outcomes; emotion-focused coping protects worker wellbeing but may delay legitimate escalation of system problems. Fourth, outcomes (Theme 4) feed back into antecedents (Theme 1) through dynamic feedback loops — when management absorbs workaround patterns as design feedback rather than as policy violations, the organisational antecedents themselves evolve, producing a learning cycle. The framework therefore treats the four themes not as parallel buckets but as an interconnected system whose loops are central to understanding digital resilience.

Building on this synthesis, the framework is articulated through six directional propositions. Crucially, these are advanced not as retrospective findings from the reviewed literature, but as conceptual propositions for future empirical testing designed to guide future research agenda. *Proposition 1a (P1a):* Human antecedents (digital literacy, computer self-efficacy, and conscientiousness) are hypothesized to be positively associated with a healthcare worker's digital resilience capacity and continuous HIS use. *Proposition 1b (P1b):* Meso-level organizational facilitators (top management support, IT helpdesk responsiveness, training, and peer networks) are hypothesized to be positively associated with a healthcare worker's digital resilience capacity and continuous HIS use. *Proposition 2 (P2):* The negative relationship between digital stressors (technostress creators) and frontline HIS continuous use is hypothesized to be moderated by technical infrastructure reliability and user-centered interface usability, such that the negative impact of threats is attenuated when technical facilitators are strong. *Proposition 3 (P3):* Digital threats and technostress creators (techno-overload, techno-invasion, techno-complexity, techno-uncertainty, techno-insecurity, security/privacy concerns) are positively associated with the deployment of coping mechanisms; the form of coping (problem-focused vs. emotion-focused vs. dynamic team capabilities) is contingent on the type of threat. *Proposition 4 (P4):* Problem-focused coping and dynamic team capabilities are positively associated with sustained continuous use, individual wellbeing, and patient-safety outcomes; emotion-focused avoidance, when used as a dominant rather than supplementary strategy, is negatively associated with these outcomes. *Proposition 5 (P5):* Resilient continuous HIS use has a hypothesized curvilinear (inverted-U) relationship with workarounds: at moderate levels, workarounds signal productive adaptation and feed back into system redesign; at high levels, they introduce systemic risks (data fragmentation, security breaches) that erode resilience — this proposition operationalises the "dark side" of resilience identified in the gaps section. *Proposition 6 (P6):* System-level outcomes (organisational agility, sustained feedback loops, scaled care models) loop back to strengthen organisational antecedents (P1), generating a learning cycle that constitutes a dynamic-capability conception of digital resilience at the institutional level. These propositions provide the basis for the future-research agenda elaborated in subsequent sections; they are intentionally generic enough to invite operationalisation across different healthcare contexts and HIS types, and specific enough to be falsifiable.

Gaps in current literature that future studies can address

To move the field beyond generic recommendations and fulfill the integrative purpose of our conceptual framework (see Fig. 3), this review establishes a prioritized future research agenda that binds our methodological, theoretical, and thematic gaps to the distinct, realities of clinical practice.

Methodologically, the current literature is bottlenecked by cross-sectional, self-reported surveys which suffer from severe social desirability bias, particularly because frontline healthcare workers are understandably hesitant to disclose unauthorized compliance violations like password sharing, shifting documentation to personal smartphones, or deploying shadow IT applications. Because a clinician cannot accurately recall or quantify these rapid workarounds during a chaotic, high-stress 12-hour shift, future studies must prioritize longitudinal, time-series designs paired with objective, time-stamped system log metadata and tracking data. This empirical shift is urgently required to rigorously test and validate the non-linear adaptation paradox hypothesized in Proposition 5, enabling researchers to mathematically pinpoint the exact threshold where localized workarounds stop being helpful, resilient shortcuts and cross the line into dangerous data fragmentation, privacy violations, and structural system errors.

Theoretically, existing research relies almost exclusively on borrowed, macro-level or individual psychological frameworks like TAM, UTAUT, and the JD-R model which treat users as isolated actors and completely ignore the intense team dynamics, rigid professional hierarchies, and ward-specific climates unique to medical environments. Future research must formulate native, meso-level sociotechnical theories to evaluate the human and organizational drivers hypothesized in P1a and P1b. This will allow researchers to model how team cohesion, multi-skilled workforce flexibility, and supervisor-worker interactions socially construct a protective, collective shield against technostress right on the clinical ward, mapping how the burden of technology manifests differently across distinct clinical roles such as an emergency room physician, a psychiatric nurse, or a home health aide.

Finally, from a technological standpoint, future thematic research must shift away from static digital documentation and focus on the implementation of advanced, probabilistic innovations like Artificial Intelligence (AI) and clinical decision support algorithms. Unlike standard database entries, AI tools generate predictive outputs that can directly conflict with a clinician's expert judgment, introducing an intense, high-stakes creator of technostress. Future inquiries should deploy qualitative action research to investigate the hidden "resilience tax"—the legal, ethical, and cognitive burden shifted entirely onto individual frontline workers who must continuously cross-examine automated outputs to protect patient safety. Investigating this dynamic is vital to validate the threat-to-coping boundaries hypothesized in P2 and P3, illuminating how technical infrastructure reliability and user-centered interface usability can be designed to actively attenuate algorithmic anxiety and secure long-term, continuous HIS use.

Conclusion

In this study, a systematic literature review combined with a grounded theory approach was conducted to synthesize the extant literature on healthcare worker digital resilience and continuous Health Information Systems (HIS) use. Guided by four research questions, the analysis consolidated the literature into four core thematic categories: antecedents and enablers, digital threats and technostress creators, coping mechanisms and resilient behaviors, and localized adaptation outcomes. Furthermore, the findings highlight a prevailing theoretical deficit, with nearly half of the corpus lacking explicit theoretical scaffolding and the remainder relying almost entirely on frameworks borrowed from general management and psychological disciplines. Methodologically, the domain exhibits a heavy reliance on cross-sectional surveys and qualitative interviews, leaving critical long-term temporal dynamics and objective utilization patterns under-researched.

While general technology acceptance research remains steady, scholarship focusing specifically on healthcare worker digital resilience and continuous HIS use has experienced a substantial expansion since 2020. This accelerating research trend is primarily driven by three factors: the rapid deployment of digital health infrastructure during recent global health crises, escalating operational concerns surrounding clinician burnout induced by chronic technostress, and the recognized imperative to ensure health information systems maintain baseline operational stability. However, this upward trajectory must be interpreted with methodological caution; because our multi-database search window concluded in early 2026, the data for the final reporting year 2026 remains partial and incomplete, representing a baseline boundary rather than a definitive decline in scholarly output. Although investigating digital resilience in high-stakes clinical environments remains a complex operational undertaking, these ongoing sociotechnical transitions present critical imperatives for researchers. By shifting focus from purely descriptive evaluations of digital stress toward the empirical testing of native, meso-level frameworks and objective log-data methodologies, future scholarship can provide the robust theoretical models necessary to support healthcare organizations during ongoing digital evolutions.

This review has several limitations that should be acknowledged. First, the search was limited to four databases (Scopus, AIS eLibrary, PubMed, IEEE Xplore) and to English-language publications; relevant work indexed elsewhere or published in other languages may have been omitted. Second, although a structured quality appraisal (MMAT for empirical studies; SANRA for reviews) was applied and an outlet-tier classification was used to weight the synthesis, some heterogeneity remains across the 60 included studies in design rigour and contextual scope. Third, the search was last executed in January 2026; given the rapid pace of publication in this area, particularly on AI-enabled HIS, recent papers may not be reflected. Fourth, although a dual-reviewer screening process with documented inter-rater reliability was employed for title-and-abstract screening and thematic coding, full text screening eligibility assessment was conducted through reviewer consensus rather than independent parallel screening, which may have introduced some risk of selection bias. Fifth, the combination of a systematic review protocol with grounded-theory coding involves a creative tension between deductive screening criteria and inductive code generation: while we addressed this by treating the coding process as theoretically open while keeping the search and appraisal protocols deductively specified, future reviews could explore this design choice more explicitly. Sixth, the six propositions advanced in the framework are conceptual rather than empirically tested; their value lies in providing falsifiable starting points for the future research agenda, and empirical validation across diverse healthcare contexts is warranted. Finally, because extensive methodological parameters including the complete search string syntaxes (Table S1), adjacent literature comparative boundaries (Table S2), and the evidence table mapping of studies with supported themes (Table S3) were relocated to supplementary materials to maintain main-text conciseness, this manuscript structurally depends on those records; their permanent availability is vital to ensure the absolute transparency, replicability, and auditable trail of the review process. These limitations notwithstanding, the review provides a structured, transparent, and rigorous synthesis of an emerging field and offers an actionable agenda for advancing scholarship on healthcare workers' digital resilience and continuous HIS use.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

CRediT authorship contribution statement

Mudasir Issah: Writing – original draft, Methodology, Data curation, Conceptualization. **Ibrahim Osman Adam:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.sbr.2026.100215.

References

Abane, A.M., Mariwah, S., Owusu, S.A., Kasim, A., Robson, E., Hampshire, K., 2021. Mobile phone use and the welfare of community health nurses in Ghana: an analysis

of unintended costs. World Dev. Perspect. 23, 100317. https://doi.org/10.1016/j.wdp.2021.100317.
Agarwal, R., Gao, G.(Gordon), DesRoches, C., Jha, A.K., 2010. The digital transformation of healthcare: current status and the road ahead. Inf. Syst. Res. 21 (4), 796–809. https://doi.org/10.1287/isre.1100.0327.
Al-Anezi, F., 2025. Challenges of healthcare systems in Saudi Arabia to delivering vision 2030: an empirical study from healthcare workers perspectives. J. Healthc. Leadersh. Volume 17, 173–187. https://doi.org/10.2147/jhl.s516159.
Alavi, M., Carlson, P., 1992. A review of MIS research and Disciplinary development. J. Manage. Inf. Syst. 8 (4), 45–62. JSTOR.
Alsyouf, A., Alsubahi, N., Alali, H., Lutfi, A., Al-Mugheed, K.A., Alrawad, M., Almaiah, M. A., Anshasi, R.J., Alhazmi, F.N., Sawhney, D., 2024. Nurses' continuance intention to use electronic health record systems: the antecedent role of personality and organisation support. PLoS. One 19 (10), e0300657. https://doi.org/10.1371/journal.pone.0300657.
Ambrose, J.W., Catchpole, K., Evans, H.L., Nemeth, L.S., Layne, D.M., Nichols, M., 2024. Healthcare team resilience during COVID-19: a qualitative study. BMC. Health Serv. Res. 24 (1), 459. https://doi.org/10.1186/s12913-024-10895-3.
Ang, W.H.D., Chew, H.S.J., Dong, J., Yi, H., Mahendren, R., Lau, Y., 2022. Digital training for building resilience: systematic review, meta-analysis, and meta-regression. Stress Health 38 (5), 848–869. https://doi.org/10.1002/smi.3154.
Argaw, S.T., Troncoso-Pastoriza, J.R., Lacey, D., Florin, M.-V., Calcavecchia, F., Anderson, D., Burleson, W., Vogel, J.-M., O'Leary, C., Eshaya-Chauvin, B., Flahault, A., 2020. Cybersecurity of Hospitals: discussing the challenges and working towards mitigating the risks. BMC. Med. Inform. Decis. Mak. 20 (1), 146. https://doi.org/10.1186/s12911-020-01161-7.
Baethge, C., Goldbeck-Wood, S., Mertens, S., 2019. SANRA—A scale for the quality assessment of narrative review articles. Res. Integr. Peer. Rev. 4 (1), 5. https://doi.org/10.1186/s41073-019-0064-8.
Bagherian, H., Sattari, M., 2022. Health information system in developing countries: a review on the challenges and causes of success and failure. Med. J. Islam. Repub. Iran. https://doi.org/10.47176/mjiri.36.111.
Beerepoot, I., Weerd, I.van de, 2018. Prevent, redesign, adopt or ignore: improving healthcare using knowledge of workarounds. Res. Papers. https://aisel.aisnet.org/ecis2018_rp/120.
Beh, L.-S., Loo, L.-H., 2012. Job stress and coping mechanisms among nursing staff in public health services. Int. J. Acad. Res. Bus. Soc. Sci. 2 (7). https://hrmars.com/papers_submitted/9164/job-stress-and-coping-mechanisms-among-nursing-staff-in-public-health-services.pdf.
Bhattacherjee, A., 2001. Understanding information systems continuance: an expectation-confirmation Model1. Manage. Inform. Syst. Quart. 25 (3), 351–370. https://doi.org/10.2307/3250921.
Boh, W., Constantinides, P., Padmanabhan, B., Viswanathan, S., 2023. Building digital resilience against major shocks. Manage. Inform. Syst. Quart. 47 (1), 343–360. https://doi.org/10.25300/MISQ/2023/47.1.12.
Bostrom, R.P., Heinen, J.S., 1977. MIS problems and failures: a socio- technical PerspectivePart I: the causes. Manage. Inform. Syst. Quart. 1 (3), 17–32. https://doi.org/10.2307/248710.
Burns, A.J., Posey, C., Roberts, T.L., Benjamin Lowry, P., 2017. Examining the relationship of organizational insiders' psychological capital with information security threat and coping appraisals. Comput. Human. Behav. 68, 190–209. https://doi.org/10.1016/j.chb.2016.11.018.
Costa, P.T., McCrae, R.R., 1992. Normal personality assessment in clinical practice: the NEO Personality Inventory. Psychol. Assess. 4 (1), 5–13. https://doi.org/10.1037/1040-3590.4.1.5.
Davis, F.D., 1989. Perceived usefulness, Perceived ease of use, and user acceptance of information technology. Manage. Inform. Syst. Quart. 13 (3), 319–340. https://doi.org/10.2307/249008.
De Vries, D.A., Piotrowski, J.T., Vreese, C.de, 2025. Developing the DigIQ: a measure of digital competence. PLoS. One 20 (5), e0322995. https://doi.org/10.1371/journal.pone.0322995.
Dehnavieh, R., Haghdoost, A., Khosravi, A., Hoseinabadi, F., Rahimi, H., Poursheikhali, A., Khajehpour, N., Khajeh, Z., Mirshekari, N., Hasani, M., Radmerikhi, S., Haghighi, H., Mehrolhassani, M.H., Kazemi, E., Aghamohamadi, S., 2019. The District Health Information System (DHIS2): a literature review and meta-synthesis of its strengths and operational challenges based on the experiences of 11 countries. Health Inform. Manage. J. 48 (2), 62–75. https://doi.org/10.1177/1833358318777713.
Delone, W.H., McLean, E.R., 2003. The DeLone and McLean model of information systems success: a ten-year update. J. Manage. Inform. Syst. 19 (4), 9–30. https://doi.org/10.1080/07421222.2003.11045748.
Doctor, E., Hennrich, J., Eymann, T., Buck, C., 2025. Understanding antecedents of nurses' and physicians' Workaround behavior regarding hospital information systems: qualitative interview study. J. Med. Internet. Res. 27, e51781. https://doi.org/10.2196/51781.
Feng, H., Shen, Y., Li, X., 2024. Bereavement coping strategies among healthcare professionals: a qualitative systematic review and meta-synthesis. Palliat. Support. Care 22, 2194–2206. https://doi.org/10.1017/S1478951524001147.
Garcia-Perez, A., Cegarra-Navarro, J.G., Sallos, M.P., Martinez-Caro, E., Chinnaswamy, A., 2023. Resilience in healthcare systems: cyber security and digital transformation. Technovation 121, 102583. https://doi.org/10.1016/j.technovation.2022.102583.
Gilson, L., Ellokor, S., Lehmann, U., Brady, L., 2020. Organizational change and everyday health system resilience: lessons from Cape Town, South Africa. Soc. Sci. Med. 266, 113407. https://doi.org/10.1016/j.socscimed.2020.113407.
Golz, C., Peter, K.A., Zwakhalen, S.M.G., Hahn, S., 2021. Technostress among health professionals – A multilevel model and group comparisons between settings and professions. Inform. Health Soc. Care 46 (2), 137–149. https://doi.org/10.1080/17538157.2021.1872579.
Gröschke, D., Hofmann, E., Müller, N.D., Wolf, J., 2022. Individual and organizational resilience—Insights from healthcare providers in Germany during the COVID-19 pandemic. Front. Psychol. 13, 965380. https://doi.org/10.3389/fpsyg.2022.965380.
Guo, Y., Luo, Y., Lam, L., Cross, W., Plummer, V., Zhang, J., 2018. Burnout and its association with resilience in nurses: a cross-sectional study. J. Clin. Nurs. 27 (1–2), 441–449. https://doi.org/10.1111/jocn.13952.
Hariyanto, H., Denison, T., Stillman, L., 2018. Understanding health information System implementation in an Indonesian Primary Health centre: a sociotechnical perspective. In: PACIS 2018 Proceedings. https://aisel.aisnet.org/pacis2018/219.
Heeks, R., Ospina, A.V., 2019. Conceptualising the link between information systems and resilience: a developing country field study. Inform. Syst. J. 29 (1), 70–96. https://doi.org/10.1111/isj.12177.
Hollnagel, E., Woods, D.D., Leveson, N. (Eds.), 2017. Resilience Engineering: Concepts and Precepts. CRC Press. https://doi.org/10.1201/9781315605685.
Hong, Q.N., Fàbregues, S., Bartlett, G., Boardman, F., Cargo, M., Dagenais, P., Gagnon, M.-P., Griffiths, F., Nicolau, B., O'Cathain, A., Rousseau, M.-C., Vedel, I., Pluye, P., 2018. The Mixed Methods Appraisal Tool (MMAT) version 2018 for information professionals and researchers. Educ. Inform. 34 (4), 285–291. https://doi.org/10.3233/EFI-180221.
Jamaluddin, A., Abawajy, J.H., 2021. Analysis of factors affecting successful adoption and acceptance of electronic health records at hospitals. In: Abawajy, J.H., Choo, K.-K.R., Chiroma, H. (Eds.), International Conference on Emerging Applications and Technologies for Industry 4.0 (EATI'2020). Springer International Publishing, pp. 282–293. https://doi.org/10.1007/978-3-030-80216-5_21.
Janssen, A., Kay, J., Talic, S., Pusic, M., Birnbaum, R.J., Cavalcanti, R., Gasevic, D., Shaw, T., 2022. Electronic health records that support health professional reflective practice: a missed opportunity in digital health. J. Healthc. Inform. Res. 6 (4), 375–384. https://doi.org/10.1007/s41666-022-00123-0.
Jeilani, A., Hussein, A., 2025. Impact of digital health technologies adoption on healthcare workers' performance and workload: perspective with DOI and TOE models. BMC. Health Serv. Res. 25 (1), 271. https://doi.org/10.1186/s12913-025-12414-4.
Jesson, J., Matheson, L., Lacey, F.M., 2011. Doing Your Literature Review: Traditional and Systematic Techniques. SAGE Publications. https://books.google.com.gh/books?id=NAYrLb8qsd4C.
Kim, D.G., Lee, C.W., 2021. Exploring the roles of self-efficacy and technical support in the relationship between techno-stress and counter-productivity. Sustainability. 13 (8), 4349. https://doi.org/10.3390/su13084349.
Kitchenham, B., 2004. Procedures For Performing Systematic Reviews, 33. Keele Univ., Keele, UK.
Kräft, J., Wirth, T., Harth, V., Mache, S., 2024. Digital stress perception among German hospital nurses and associations with health-oriented leadership, emotional exhaustion and work-privacy conflict: a cross-sectional study. BMC. Nurs. 23 (1), 213. https://doi.org/10.1186/s12912-024-01825-z.
Kutney-Lee, A., Sloane, D., Bowles, K., Burns, L., Aiken, L., 2019. Electronic health record adoption and nurse reports of usability and quality of care: the role of work environment. Appl. Clin. Inform. 10 (01), 129–139. https://doi.org/10.1055/s-0039-1678551.
Larsson, M., Ho, D.M., Kirschner, M., Seifritz, E., Manoliu, A., 2025. Digital resilience interventions for healthcare workers: a systematic review. Front. Psychiatry 16, 1519670. https://doi.org/10.3389/fpsyt.2025.1519670.
Li, F., Ma, Q., Yang, C., Zhong, M., 2025. Investigating key elements of digital resilience among nursing undergraduates: a qualitative study. Front. Med. (Lausanne) 11, 1452580. https://doi.org/10.3389/fmed.2024.1452580.
Liu, Y., Xu, X., Jin, Y., Deng, H., 2023. Understanding the digital resilience of physicians during the COVID-19 Pandemic: an empirical study. MIS Quarterly 47 (1), 391–422. https://doi.org/10.25300/misq/2022/17248.
Luthans, F., 2002. The need for and meaning of positive organizational behavior. J. Organ. Behav. 23 (6), 695–706. https://doi.org/10.1002/job.165.
Maraj, M.A.A., Hossain, M.A., Islam, S., Arif, Nur Uddin M., 2024. Information systems in health management: innovations and challenges in the digital era. GLOBAL Mainstream J. 1 (2), 14–25. https://doi.org/10.62304/ijhm.v1i2.128.
Mariwah, S., Machistey Abane, A., Asiedu Owusu, S., Kasim, A., Robson, E., Castelli, M., Hampshire, K., 2022. Formalising 'informal' mHealth in Ghana: opportunities and challenges for Universal Health coverage (UHC). Glob. Public Health 17 (5), 768–781. https://doi.org/10.1080/17441692.2021.1874467.
Mensah, N.K., Adzakpah, G., Kissi, J., Boadu, R.O., Acheampongmaa, Q.K., Taylor-Abdulai, H., Chatio, S.T., 2023. Perceived impact of digital health technology on health professionals and their work: a qualitative study in Southern Ghana. Digit. Health 9. https://doi.org/10.1177/20552076231218838, 20552076231218838.
Mo, Y., Zhang, H., Bai, X., Lei, S., Han, L., 2025. Digital resilience of clinical nurses: a concept analysis. BMC. Nurs. 24 (1), 1231. https://doi.org/10.1186/s12912-025-03713-6.
Okronipa, A.Q., Nyame, J.Y., 2024. Adoption of health information systems in emerging economies: evidence from Ghana. The Sci. Temper 15 (03), 2553–2563. https://doi.org/10.58414/SCIENTIFICTEMPER.2024.15.3.22.
Oprea, N., Giacomelli, G., Sartirana, M., Trinchero, E., Georgescu, I., 2025. Meso-organisational determinants of healthcare workers' resilience: results of a scoping review. Health Policy. (New. York) 161, 105412. https://doi.org/10.1016/j.healthpol.2025.105412.

Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., Shamseer, L., Tetzlaff, J.M., Akl, E.A., Brennan, S.E., Chou, R., Glanville, J., Grimshaw, J.M., Hróbjartsson, A., Lalu, M.M., Li, T., Loder, E.W., Mayo-Wilson, E., McDonald, S., Moher, D., 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 372, n71. https://doi.org/10.1136/bmj.n71 n71.

Phan, T.A., Le, T.M.T., Pham, B.T.T., 2026a. Live influence: redefining credibility and attractiveness in healthcare livestreaming. Howard J. Commun. 37 (1), 210–234. https://doi.org/10.1080/10646175.2024.2429454.

Phan, T.A., Nguyen, P.N.Q., Le, T.M.T., Pham, B.T.T., 2026b. Digital dilemmas: the complex interplay of online experience and health pharmaceutical purchases among Vietnamese women. Howard J. Commun. 37 (2), 378–398. https://doi.org/10.1080/10646175.2025.2458014.

Pilosof, N.P., Welcman, Y., Barrett, M., Oborn, E., Barrett, S., 2025. Building digital resilience: leading healthcare transformation through an online community. Front. Digit. Health 7, 1656804. https://doi.org/10.3389/fdgth.2025.1656804.

Pirkkalainen, H., Salo, M., Tarafdar, M., Makkonen, M., 2019. Deliberate or instinctive? Proactive and reactive coping for technostress. J. Manage.Inform. Syst. 36 (4), 1179–1212. https://doi.org/10.1080/07421222.2019.1661092.

Provenzano, M., Cillara, N., Curcio, F., Pisu, M.O., González, C.I.A., Jiménez-Herrera, M. F., 2024. Electronic health record adoption and its effects on healthcare staff: a qualitative study of well-being and workplace stress. Int. J. Environ. Res. Public Health 21 (11), 1430. https://doi.org/10.3390/ijerph21111430.

Ragu-Nathan, T.S., Tarafdar, M., Ragu-Nathan, B.S., Tu, Q., 2008. The consequences of technostress for end users in organizations: conceptual development and empirical validation. Inform. Syst. Res. 19 (4), 417–433. https://doi.org/10.1287/isre.1070.0165.

Rahman, S., Islam, A., 2024. Healthcare information systems (HIS): implementation challenges in developing countries. Bangl. J. Med. Sci. 23 (2), 314–326. https://doi.org/10.3329/bjms.v23i2.72094 (Australia; Malaysia).

Rubbio, I., Bruccoleri, M., Pietrosi, A., Ragonese, B., 2019. Digital health technology enhances resilient behaviour: evidence from the ward. Int. J. Oper. Prod. Manage. 40 (1), 34–67. https://doi.org/10.1108/IJOPM-02-2018-0057.

Salifu Sidii, F., 2024. Navigating the intersection of digital security, resilience and sustainability in healthcare: a theoretical framework and case study of Ghana. Health Econ. Manage. Rev. 5 (4), 130–146. https://doi.org/10.61093/hem.2024.4-09.

Sauer, P.C., Seuring, S., 2023. How to conduct systematic literature reviews in management research: a guide in 6 steps and 14 decisions. Rev. Manager. Sci. 17 (5), 1899–1933. https://doi.org/10.1007/s11846-023-00668-3.

Shan, Y., Shang, J., Yan, Y., Ye, X., 2023. Workflow interruption and nurses' mental workload in electronic health record tasks: an observational study. BMC. Nurs. 22 (1), 63. https://doi.org/10.1186/s12912-023-01209-9.

Simbini, T., Adimado, E., Adjorlolo, S., Guerrero-Torres, L., Srinivas, P., Zizhou, S., Zerfu, T., 2026. Digital health interventions in strengthening primary healthcare systems in Sub-Saharan Africa: insights from Ethiopia, Ghana, and Zimbabwe. PLOS. Digit. Health 5 (1), e0000863. https://doi.org/10.1371/journal.pdig.0000863.

Sittig, D.F., Singh, H., 2010. A new sociotechnical model for studying health information technology in complex adaptive healthcare systems. BMJ Qual. Saf. 19 (Suppl 3), i68–i74. https://doi.org/10.1136/qshc.2010.042085.

Sun, H., Yuan, C., Qian, Q., He, S., Luo, Q., 2022. Digital resilience among individuals in school education settings: a concept analysis based on a scoping review. Front. Psychiatry 13, 13–2022. https://doi.org/10.3389/fpsyt.2022.858515.

Tahara, M., Mashizume, Y., Takahashi, K., 2020. Coping mechanisms: exploring strategies utilized by Japanese healthcare workers to reduce stress and improve mental health during the COVID-19 pandemic. Int. J. Environ. Res. Public Health 18 (1), 131. https://doi.org/10.3390/ijerph18010131.

Tarafdar, M., Pullins, E.B., Ragu-Nathan, T.S, 2015. Technostress: negative effect on performance and possible mitigations. Inform. Syst. J. 25 (2), 103–132. https://doi.org/10.1111/isj.12042.

Tranfield, D., Denyer, D., Smart, P., 2003. Towards a methodology for developing evidence-informed management knowledge by means of systematic review. Br. J. Manage. 14 (3), 207–222. https://doi.org/10.1111/1467-8551.00375.

Tzitzili, G.A., Spyrou, S., Platis, A.N., 2023. An analysis of resilience and risk assessment in Digital Healthcare systems. In: Proceeding of the 33rd European Safety and Reliability Conference, pp. 599–606. https://doi.org/10.3850/978-981-18-8071-1_P420-cd.

Venkatesh, Morris, Davis, Davis, 2003. User acceptance of information technology: toward a unified view. MIS Quart. 27 (3), 425–478. https://doi.org/10.2307/30036540.

Wallace, S., 2017. Effect of EMR use on technostress and healthcare providers. In: MWAIS 2017 Proceedings. https://aisel.aisnet.org/mwais2017/5.

Webster, J., Watson, R.T., 2002. Analyzing the past to prepare for the future: writing a literature review. Manage. Inform. Syst. Quart. 26 (2), xiii–xxiii. https://doi.org/10.2307/4132319.

Wirkkala, M., Wijk, K., Larsson, A.C., Engström, M., 2024. Technology frustration in healthcare – does it matter in staff ratings of stress, emotional exhaustion, and satisfaction with care? A cross-sectional correlational study using the job demands-resources theory. BMC. Health Serv. Res. 24 (1), 1557. https://doi.org/10.1186/s12913-024-11906-z.

Woldemariam, M.T., Jimma, W., 2023. Adoption of electronic health record systems to enhance the quality of healthcare in low-income countries: a systematic review. BMJ Health Care Inform. 30 (1), e100704. https://doi.org/10.1136/bmjhci-2022-100704.

Wolfswinkel, J.F., Furtmueller, E., Wilderom, C.P.M., 2013. Using grounded theory as a method for rigorously reviewing literature. Eur. J. Inform. Syst. 22 (1), 45–55. https://doi.org/10.1057/ejis.2011.51.

Ye, D., Xu, B., Wei, B., Zheng, L., Wu, Y.J., 2024. Employee work engagement in the digital transformation of enterprises: a fuzzy-set qualitative comparative analysis. Human. Soc. Sci. Commun. 11 (1), 35. https://doi.org/10.1057/s41599-023-02418-y.

Zheng, Y., Huang, W., Wang, J., Wu, X., Xu, Y., 2016. *An empirical investigation into the effect of technostress of physicians on adoption of electronic healthcare systems.* https://aisel.aisnet.org/pacis2016/388.